AP SBTET — C-23

Diploma in Computer Engineering (CME)

CM-501
Industrial Management and Entrepreneurship

Lesson-wise Question Bank with Answers

Included for every unit: MCQs with four options and answers, short-answer questions with model answers, and long-answer questions with worked explanations or examples.

Scope note: This is a comprehensive original practice bank aligned to the topics and learning outcomes listed in the AP SBTET C-23 CME curriculum. It is not an official SBTET answer key, and no question bank can guarantee every possible exam question. Use it alongside your classroom notes and official syllabus.

Contents and revision method

Study the unit headings in order. Attempt MCQs before checking the answer, practise short answers in your own words, and structure long answers with definitions, headings, steps, examples and diagrams where suitable.

1. Principles of Management

Topics covered: Meaning, nature and functions of management; Fayol's principles; Management levels and skills; Planning, organizing, staffing, directing and controlling

A. Multiple-Choice Questions

1. Management is best described as:
A. A process of achieving goals through people and resources
B. Only bookkeeping
C. Only manufacturing
D. Only advertising
Answer: A. A process of achieving goals through people and resources

2. Which function sets objectives and decides future action?
A. Controlling
B. Planning
C. Staffing
D. Directing
Answer: B. Planning

3. Which skill is especially important for top-level management?
A. Conceptual skill
B. Typing speed
C. Machine operation only
D. Data entry only
Answer: A. Conceptual skill

B. Short-Answer Questions

1. Define management.

Answer: Management is the process of planning, organizing, staffing, directing and controlling resources so that organizational objectives are achieved efficiently and effectively.

2. List the major functions of management.

Answer: The main functions are planning, organizing, staffing, directing or leading, coordinating and controlling.

3. What are the three levels of management?

Answer: Top-level management sets overall policy and long-term goals; middle-level management converts policy into departmental plans; lower-level or supervisory management guides day-to-day work.

C. Long-Answer Questions

1. Explain Fayol's principles of management.

Answer: Henri Fayol proposed principles such as division of work, authority with responsibility, discipline, unity of command, unity of direction, subordination of individual interest, fair remuneration, centralization balance, scalar chain, order, equity, stability of tenure, initiative and team spirit. These principles help managers coordinate people and work.

2. Explain the functions of management with examples.

Answer: Planning defines objectives and methods. Organizing groups tasks and assigns authority. Staffing recruits and trains employees. Directing communicates, motivates and supervises. Coordinating aligns departments. Controlling compares actual performance with standards and corrects deviations. For example, a factory plans output, allocates machines and workers, trains staff, supervises production and checks quality against targets.

2. Organization Structure and Organizational Behaviour

Topics covered: Line, functional, line-and-staff and committee structures; Recruitment, selection, training and performance appraisal; Leadership theories and styles; Motivation, communication and human relations

A. Multiple-Choice Questions

1. In a line organization, authority usually flows:
A. Through a clear vertical chain
B. Only between customers
C. Without any supervisor
D. Only through outside consultants
Answer: A. Through a clear vertical chain

2. Maslow's theory is about:
A. Hierarchy of human needs
B. Inventory valuation
C. Product routing
D. Cloud storage
Answer: A. Hierarchy of human needs

3. Selection tests are used mainly to:
A. Assess suitability of applicants
B. Set product prices
C. Repair machinery
D. Calculate depreciation
Answer: A. Assess suitability of applicants

B. Short-Answer Questions

1. What is organizational structure?

Answer: Organizational structure is the formal arrangement of jobs, authority, responsibilities and communication within an organization.

2. Differentiate recruitment and selection.

Answer: Recruitment attracts eligible applicants to apply for a job. Selection evaluates applicants and chooses the most suitable person.

3. Define leadership and motivation.

Answer: Leadership is the ability to influence people toward shared goals. Motivation is the set of internal and external factors that encourage people to act and sustain effort.

C. Long-Answer Questions

1. Explain line and staff organization with advantages and limitations.

Answer: A line organization has a direct chain of command and clear responsibility. A staff function provides specialist advice to line managers. Advantages include clear authority and expert support. Limitations may include conflict between line and staff, slower decisions and extra administrative cost.

2. Explain Maslow's hierarchy of needs and its use at work.

Answer: Maslow grouped needs into physiological, safety, social, esteem and self-actualization levels. In a workplace, fair pay supports basic needs, safe conditions support security, teamwork supports belonging, recognition supports esteem, and challenging work supports personal growth. People may have several needs at once, so managers should not treat the hierarchy as a rigid rule.

3. Production Management

Topics covered: Production types and production planning; Routing, scheduling, dispatching and follow-up; Inventory control and stores management; Quality control, TQM, break-even analysis and project networks

A. Multiple-Choice Questions

1. Routing determines:
A. The path and sequence of operations
B. Only salary rates

C. Only advertising channels
D. Only the final invoice
Answer: A. The path and sequence of operations

2. TQM stands for:
A. Total Quality Management
B. Technical Quantity Method
C. Total Queue Monitoring
D. Temporary Quality Measure
Answer: A. Total Quality Management

3. Break-even point is where:
A. Total revenue equals total cost
B. Profit is always maximum
C. Fixed cost is zero
D. Sales are zero
Answer: A. Total revenue equals total cost

B. Short-Answer Questions

1. What is inventory control?

Answer: Inventory control is the systematic regulation of stock levels so materials are available when needed while avoiding excessive stock and carrying costs.

2. Differentiate routing and scheduling.

Answer: Routing determines where and in what sequence work is performed. Scheduling determines when each operation should start and finish.

3. What is Total Quality Management?

Answer: TQM is an organization-wide approach focused on customer satisfaction, continuous improvement, employee involvement and prevention of defects.

C. Long-Answer Questions

1. Explain production planning and control.

Answer: Production planning decides what to produce, how to produce it, what materials and capacity are needed, and when work should occur. Control includes dispatching work, monitoring progress, inspecting output and correcting delays or quality problems. It improves delivery reliability, resource use and cost control.

2. Explain break-even analysis and its importance.

Answer: The break-even point is the output or sales level where total revenue equals total cost, so profit is zero. In units, BEP = Fixed Costs ÷ (Selling Price per Unit – Variable Cost per Unit). It helps estimate minimum sales, compare pricing options and understand risk. A break-even chart plots cost and revenue against output.

4. Engineering Ethics, Safety and Labour Codes

Topics covered: Engineering ethics and human values; Industrial accidents, causes, costs and prevention; Occupational safety, health and working conditions; Industrial Relations Code and labour-law awareness; ISO 9000 and TQM

A. Multiple-Choice Questions

1. PPE means:
A. Personal Protective Equipment
B. Production Planning Estimate
C. Public Process Entry
D. Personal Performance Evaluation
Answer: A. Personal Protective Equipment

2. ISO 9000 standards relate mainly to:
A. Quality management systems
B. Mobile phone generations
C. Programming syntax

D. Wireless frequencies
Answer: A. Quality management systems

3. A common cause of industrial accidents is:
A. Unsafe acts or unsafe conditions
B. Proper machine guarding
C. Regular training
D. Good housekeeping
Answer: A. Unsafe acts or unsafe conditions

B. Short-Answer Questions

1. What is engineering ethics?

Answer: Engineering ethics applies values such as honesty, competence, fairness, accountability and protection of public safety to engineering decisions and professional conduct.

2. List common causes of industrial accidents.

Answer: Causes include inadequate training, unsafe machinery, missing guards, poor housekeeping, fatigue, improper PPE, faulty procedures, electrical hazards and failure to report risks.

3. State the benefits of ISO 9000 certification.

Answer: It can improve consistency, documentation, customer confidence, process control, corrective action and access to markets that expect a quality management system.

C. Long-Answer Questions

1. Explain industrial accident causes, costs and prevention.

Answer: Accidents may result from unsafe acts, unsafe conditions, poor training, defective equipment or weak supervision. Costs include injury, medical expenses, lost work time, damaged equipment, investigation, compensation and reputational harm. Prevention requires hazard identification, risk assessment, guards, maintenance, PPE, training, emergency procedures and regular audits.

2. Explain the importance of workplace safety and ethical conduct.

Answer: Safety protects workers, visitors, equipment and the public. Employers should identify hazards, provide safe equipment, train workers and investigate incidents. Engineers should report defects truthfully, follow standards, protect confidential information and avoid decisions that put public welfare at risk. Labour codes establish legal expectations for working conditions, safety and industrial relations; organizations should follow current applicable rules.

5. Entrepreneurship and Start-ups

Topics covered: Entrepreneurship and qualities of entrepreneurs; Business opportunity, feasibility and business plans; Sources of finance and government support/EDP schemes; Start-up development, marketing, risk and project planning

A. Multiple-Choice Questions

1. An entrepreneur primarily:
A. Identifies opportunities and organizes a venture
B. Avoids every decision
C. Only buys products for personal use
D. Never accepts responsibility
Answer: A. Identifies opportunities and organizes a venture

2. EDP stands for:
A. Entrepreneurship Development Programme
B. Electronic Data Protocol
C. Engineering Design Process only
D. Employee Duty Plan
Answer: A. Entrepreneurship Development Programme

3. A business plan should include:
A. Market, operations and financial estimates
B. Only the logo

C. Only the owner's name
D. No cost information
Answer: A. Market, operations and financial estimates

B. Short-Answer Questions

1. Define entrepreneur and entrepreneurship.

Answer: An entrepreneur identifies a business opportunity, organizes resources, makes decisions and accepts calculated risk. Entrepreneurship is the process of developing and operating a venture to create value.

2. What is a feasibility study?

Answer: A feasibility study evaluates whether a business idea is practical by examining market demand, technical requirements, operating needs, legal factors and financial viability.

3. List common sources of finance for a small enterprise.

Answer: Sources may include personal savings, bank loans, eligible government schemes, microfinance, partners, investors and retained earnings. The suitable source depends on eligibility, cost, risk and repayment ability.

C. Long-Answer Questions

1. Explain the steps involved in starting a small enterprise.

Answer: Identify a customer problem, research demand and competitors, assess technical and financial feasibility, prepare a business plan, select the business structure, arrange funds, complete applicable registrations, obtain resources, launch marketing, begin operations and monitor cash flow and customer feedback.

2. Explain the contents and importance of a business plan.

Answer: A business plan describes the idea, product or service, target market, competition, marketing approach, operations, staffing, legal requirements, funding needs, projected costs and revenue, risks and milestones. It helps test assumptions, guide decisions and communicate the proposal to lenders or partners.

Syllabus reference: AP SBTET Curriculum-2023 (C-23), Diploma in Computer Engineering, Semester V. This study aid paraphrases syllabus topics and provides original practice questions and model answers.